Python Programmation

CM 4 : First Course Recap

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During this recap, we will correct exercises from previous practical sessions.

Short exercises: using the board.
Long exercises: by group (10 groups max).

CM1 Doggy bag

Python Basics

- Variables and types
- Operators
- Console input/output
- File manipulation

Control Flow

- Statements
- Loops



CM2 Doggy bag

Data structures

- Lists
- Dictionaries
- Tuples
- Sets

Advanced Python Stuffs

- Advanced loops
- Functions



CM3 Doggy bag

Functions

- Function definition and calling
- Parameters and arguments
- Variadic arguments
- Python scopes

Advanced strings

- Special characters handling
- String formatting



Before we start, here are some common mistakes
I saw in your notebooks.

Respect the Rules

- Here is the intented name of a notebook: TD1_LASTNAME_LASTNAME.ipynb
- A TD without your name ⇒ points denied
- A TD without the name of your binom ⇒ points denied

Do What I Ask for

- "Write a function to..." ⇒ function expected

```
a = 1
b = 2

my_add_function = ( a + b)
```

Identation Error

- Stick to one identation scheme (2 spaces or 4 spaces for instance, not both)

```
def my_add_function(a,b):
    return a + b

def my_add_function(a,b):
  return a + b
```

Identation Error II, the Return

- "return" should not be in a loop, unless you add a condition

```
def find_mystery_number():
    guess = 1
    while hot_or_cold(guess) != "bingo bango!":
        guess += 1
        return guess
```

Variables Names Matters

- Don't use variable names which are used in Python builds-in

```
list = []
list.append(i)
len(list)
list[i]
```

Function Calling

- Be careful with function calling, names must be respected

```
def convert_fahr_to_cels(def_fahr):
    # do stuff...

fahrenheit_to_celsius(fahrenheit)
```

"Inspiration" VS plagiarism

- DO NOT COPY/PASTE SOMEONE ELSE CODE !!!

```
# Influenced by "some url"

def collatz_number(n):
    # do stuff...
```

Other Remarks

- Always make sure that you call your functions or print the result in a cell
- If I have to edit your notebook to see a cell output, I will substract some points
- Be careful when sending yout TDs, I got two empty TDs...

Now, let's proceed to the corrections !

Infinite Loop

```
def find_mystery_number():
    guess = 10
    while True:
        result = hot_or_cold(guess)
        if result == "bingo bango!":
            return guess
        if result == "you're hot!":
            break
        guess += 9
    guess -= 4
    while True:
        result = hot_or_cold(guess)
        if result == "bingo bango!":
            return guess
        guess += 1
```

String Warning

```
# write your code here
dict_contents=({"chimps": 42, "peanuts": 0})

def silly_messager(**d):
print(f"As for chimps, I've got {d["chimps"]}, sadly.")

silly_messager(**dict_contents)
```